

MODULE 19

SQL NULL Handling and CASE Expressions

DBMS with SQL - Enterprise Learning Series

ARSHITH LMS | Professional Edition

SQL NULL Handling and CASE Expressions

Understanding NULL

NULL = unknown or missing value.

NOT zero, NOT empty string.

NULL = NULL is FALSE

NULL != NULL is FALSE

Any math with NULL = NULL

Use IS NULL, never = NULL

IS NULL Queries

```
SELECT * FROM students WHERE email IS NULL;
SELECT * FROM students WHERE email IS NOT NULL;

-- Count missing emails
SELECT COUNT(*) - COUNT(email) AS no_email FROM students;
```

COALESCE and IFNULL

```
COALESCE(NULL, NULL, 'default') -- returns 'default'

SELECT first_name,
COALESCE(phone, email, 'No contact') AS contact
FROM students;

IFNULL(phone, 'N/A') -- NULL becomes N/A
NULLIF(gpa, 0) -- returns NULL if gpa = 0
```

CASE Expression

```
SELECT first_name,
CASE
WHEN gpa >= 3.7 THEN 'A+ Distinction'
WHEN gpa >= 3.3 THEN 'A Excellent'
WHEN gpa >= 3.0 THEN 'B+ Good'
ELSE 'B Average'
END AS grade_label
FROM students;
```

CASE in Aggregates

```
SELECT
COUNT(CASE WHEN gpa >= 3.5 THEN 1 END) AS honor_roll,
COUNT(CASE WHEN gpa < 2.0 THEN 1 END) AS at_risk,
```

```
COUNT(*) AS total  
FROM students;
```